



Understanding PCOS

Polycystic Ovary Syndrome

A comprehensive analysis of 541 patients revealing patterns, prevalence, and pathways to better health outcomes.



PCOS at a Glance

32.7%

PCOS Prevalence

Among 541 patients
analyzed

24

Average BMI

Across all patient groups

25-29

Peak Age Range

Most affected age group
in years

64

**Cases in Peak
Range**

Highest concentration of
patients

The most common symptom reported is pimples, affecting a significant portion of patients. Age distribution shows declining cases after 29, with only 9 cases in patients 40 and above.



BMI and Lifestyle Patterns

BMI Distribution

Most PCOS patients are overweight, healthy, or obese. Non-PCOS patients tend to be healthy or underweight.

BMI remains consistent across age groups, with 25-29 and 35-39 showing slightly higher averages.

Lifestyle Impact

Higher PCOS cases among those who:

- Do not exercise regularly
- Consume fast food frequently
- Maintain sedentary habits

Mild PCOS

Early intervention opportunities

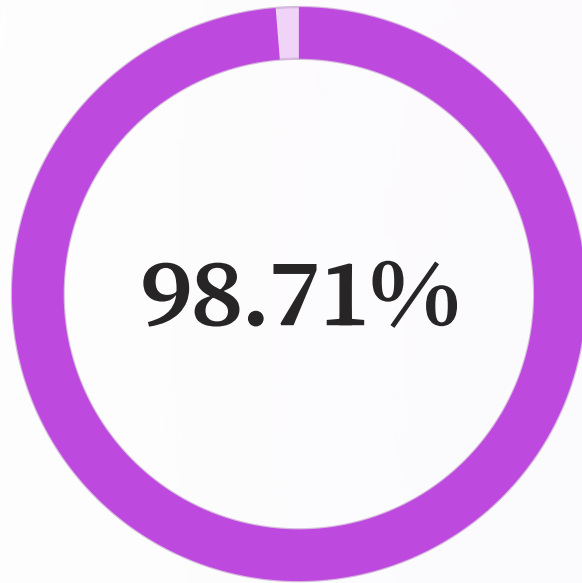
Moderate PCOS

Requires lifestyle modification

Severe PCOS

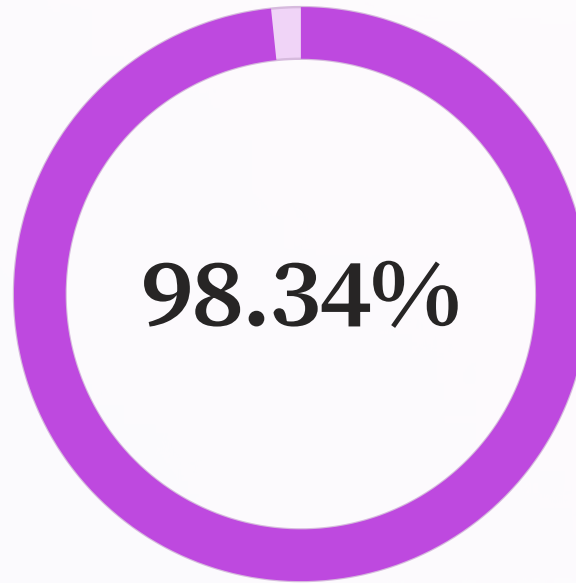
Comprehensive treatment needed

Hormonal & Metabolic Profile



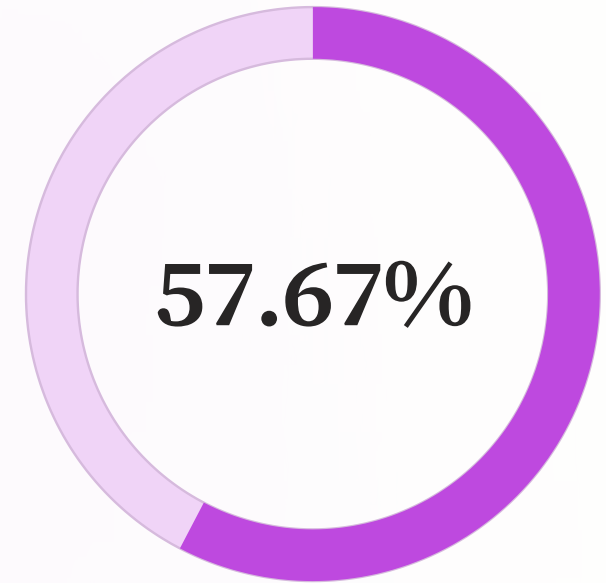
High Glucose Risk

Among PCOS patients



Diabetes Prevalence

Elevated metabolic concern



Hormonal Imbalance

In PCOS group

Key Hormone Levels

- Average TSH: 3 mIU/L
- Average AMH: 6 ng/mL overall
- PCOS AMH: 7.84 ng/mL (elevated)
- PCOS Prolactin: 24.43 ng/mL

Thyroid Status

80% normal thyroid function, 19% hypothyroid, 1% hyperthyroid across both groups.

Status varies with age, showing peaks in certain clusters.



Metabolic Risk Factors

Glucose status across BMI and thyroid metrics emphasizes elevated metabolic risk in PCOS, reinforcing the critical need for early intervention and lifestyle modification.



High Glucose Risk

Nearly universal among PCOS patients



Diabetes Prevalence

Affects 98% of PCOS group



Early Intervention

Essential for prevention

Menstrual & Reproductive Health



Follicle Counts

PCOS: 12.7 average vs. Controls: 8.6 average. Aligns with diagnostic criteria of 12+ antral follicles.



Cycle Irregularities

Up to 80% experience oligomenorrhea or amenorrhea due to disrupted ovulation.



Fertility Challenges

70-80% anovulation rate. PCOS accounts for 70-80% of anovulatory infertility cases.

Endometriosis prevalence at 4.9% with hormonal imbalance at 57.76%, consistent with 2-5% overlap studies. Pregnancy metrics show higher rates in PCOS (20.6), underscoring fertility challenges.

Clinical Symptoms & Diagnostics

Dominant Symptoms

Pimples (Acne): Affects 15-40% due to androgen excess

Hair Loss: Androgenic alopecia or hirsutism in 70-80% of cases

Severity levels range from moderate to high, correlating with hyperandrogenism affecting skin and hair follicles via DHT conversion.

Vital Signs

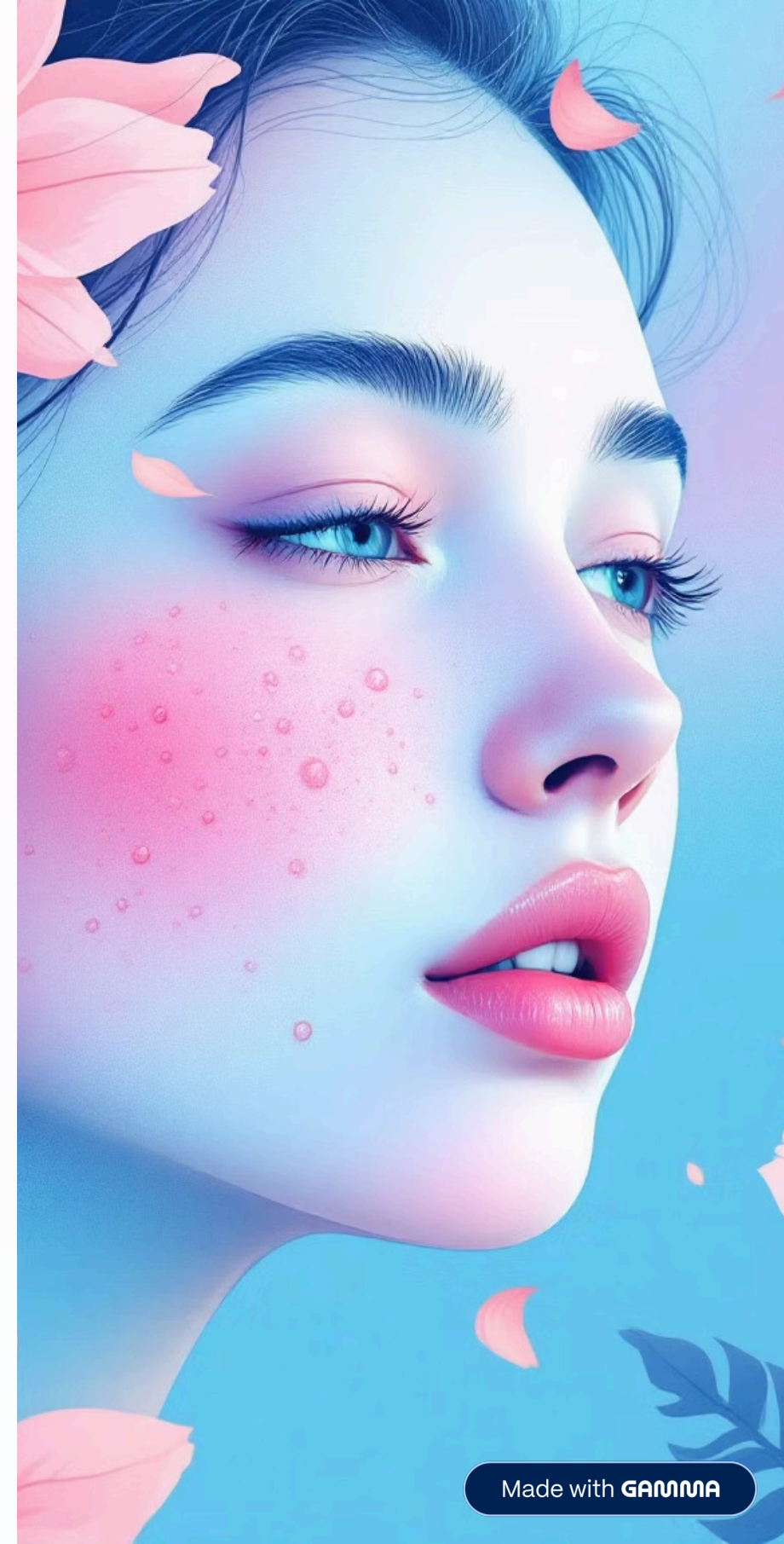
- Average Pulse: 73 BPM
- Blood Pressure: 73/53 mmHg
- Hypertension risk: 40% higher

Ultrasound Findings

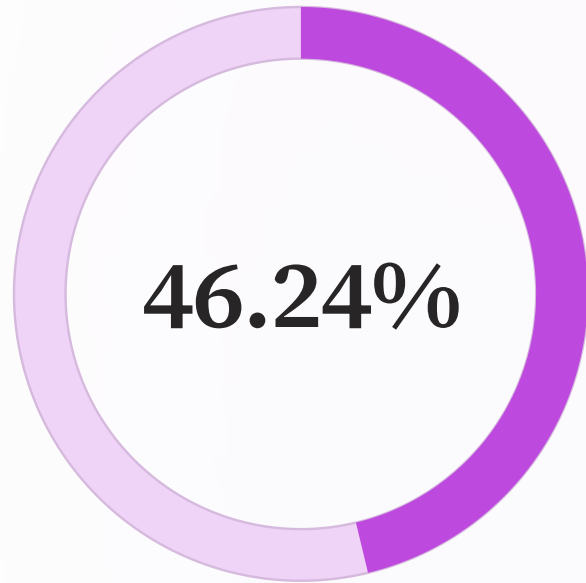
String-of-pearls pattern diagnostic in 80% when combined with clinical features.

Weight Patterns

50-70% of PCOS women are overweight, exacerbating symptoms like central adiposity.

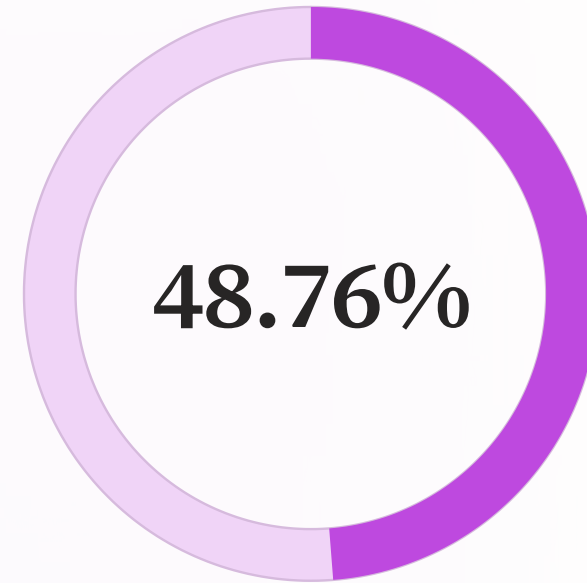


Exercise & Wellness Connection



No Exercise

Among PCOS patients



Regular Exercise

Shows protective effect

Moderate aerobic or strength training 150-300 minutes per week improves insulin sensitivity and ovulation rates in 50-70% of PCOS cases.

Menstrual Duration

Average: 4.9 days (normal 3-7 range). PCOS cycles often extend 32-35 days due to anovulation.

Waist Measurement

Average: 81 cm, reflecting central obesity in 60-80% of PCOS women, heightening cardiovascular risk.

Diet, Blood Groups & Pregnancy



Fast Food Impact

Higher PCOS prevalence with junk food consumption. High carbs/fats increase obesity and acne in 60-80% via insulin resistance.



Blood Group Patterns

O positive shows higher association in some studies (30-40% prevalence), though ABO/Rh links remain inconsistent.



Pregnancy Outcomes

20-40% lower natural rates, but 45% IVF success. Increased miscarriage (25-30%) and preterm risks.



Key Takeaways & Action Steps

01

Early Detection Matters

Peak age 25-29 with 32.72% prevalence. Monitor symptoms like acne and irregular cycles early.

02

Metabolic Health is Critical

98%+ glucose risk and diabetes prevalence demand immediate lifestyle intervention.

03

Exercise & Diet Transform Outcomes

150-300 min/week exercise improves ovulation in 50-70%. Avoid fast food to reduce insulin resistance.

04

Fertility Support Available

Despite 70-80% anovulation, IVF shows 45% success. Early intervention improves natural conception rates by 20-50%.